NATIONAL BOARD OF ACCREDITATION

Data Capturing Points of the Program Applied for NBA Accreditation– Tier I/II UG (Engineering) Institute Programs

Program Name : Biomedical Engineering	**Discipline**: Engineering & Technology
Level : Under Graduate	**Tier**: 1
Application No: 11969	**Date of Submission**: 18-07-2026

PART A- Profile of the Institute

A1.Name of the Institute: Sri Shakthi Instituteof Engineering and Technology	
Year of Establishment : 2006	Location of the Institute: Coimbatore
A2. Institute Address:Sri Shakthi Nagar, L& T By-pass, Chinniampalayam (Post) Coimbatore - 641 062	
City:Coimbatore	State:Tamil Nadu
Pin Code:641062	Website:www.siet.ac.in
Email:principal@siet.ac.in	Phone No(with STD Code):0422-2369900
A3. Name and Address of the Affiliating University (if any):	
Name of the University : Anna University	City: Coimbotore
State : Tamil Nadu	Pin Code: 600025
A4. Type of the Institution: Any Other(Please Specify)	
A5. Ownership Status: Self financing	

A6. Details of all Programs being Offered by the Institution:

- No. of UG programs: **14**
- No. of PG programs: **7**

Table No. A6.1: List of all programs offered by the Institute.

Sr.No.	Discipline	Level of program	Name of the program	Year of Start	Year of Closed	Name of The Department
1	Engineering & Technology	UG	Agricultural Engineering	2015	--	Agricultural Engineering
2	Engineering & Technology	UG	Artificial Intelligence and Data Science	2021	--	Artificial Intelligence and Data Science
3	Engineering & Technology	UG	Artificial Intelligence and Machine Learning	2021	--	Artificial Intelligence and Machine Learning
4	Engineering & Technology	UG	Biomedical Engineering	2016	--	Biomedical Engineering
5	Engineering & Technology	UG	Biotechnology	2016	--	Biotechnology
6	Engineering & Technology	PG	CAD/CAM	2013	--	Mechanical Engineering
7	Engineering & Technology	UG	Civil Engineering	2011	--	Civil Engineering
8	Engineering & Technology	UG	Computer Science and Engineering	2006	--	Computer Science and Engineering
9	Engineering & Technology	PG	Computer Science and Engineering	2010	--	Computer Science and Engineering
10	Engineering & Technology	UG	Computer Science and Engineering (Cyber Security)	2022	--	Computer Science and Engineering (Cyber Security)

11	Engineering & Technology	UG	Electrical and Electronics Engineering	2006	--	Electrical and Electronics Engineering
12	Engineering & Technology	UG	Electronics & Communication Engineering	2006	--	Electronics and Communication Engineering
13	Engineering & Technology	UG	Electronics Engineering (VLSI Design and Technology)	2024	--	Electronics Engineering (VLSI Design and Technology)
14	Engineering & Technology	PG	Embedded Systems Technology	2011	--	Electrical and Electronics Engineering
15	Engineering & Technology	PG	Farm Machinery	2024	--	Agricultural Engineering
16	Engineering & Technology	UG	Food Technology	2016	--	Food Technology
17	Engineering & Technology	PG	Food Technology	2024	--	Food Technology
18	Engineering & Technology	UG	Information Technology	2006	--	Information Technology
19	Engineering & Technology	UG	Mechanical Engineering	2009	--	Mechanical Engineering
20	Engineering & Technology	PG	Structural Engineering	2014	--	Civil Engineering
21	Engineering & Technology	PG	VLSI Design	2012	--	Electronics and Communication Engineering

A7. Programs to be considered for Accreditation vide this Application:

Table No. A7.1: List of programs to be considered for accreditation.

Name of the Department	Having Allied Departments	Name of the Program	Program Level
Civil Engineering	No	Civil Engineering	UG
Biotechnology	No	Biotechnology	UG
Biomedical Engineering	Yes	Biomedical Engineering	UG
Artificial Intelligence and Machine Learning	Yes	Artificial Intelligence and Machine Learning	UG
Artificial Intelligence and Data Science	Yes	Artificial Intelligence and Data Science	UG

Table No. A7.2: Allied Department(s) to the Department of the program considered for accreditation as above.
Cluster ID. Name of the Department (in table no. A7.1) Name of allied Departments/Cluster (for table no. A7.1)

Allied Department/Cluster Name	Program Name	Program Level
Electronics and Communication Engineering	Electronics & Communication Engineering	UG
Electronics Engineering (VLSI Design and Technology)	Electronics Engineering (VLSI Design and Technology)	UG
Electronics and Communication Engineering	VLSI Design	PG

PART-B: Program information

B1. Provide the Required Information for the Program Applied For:

Table No. B1: Program details.
A. List of the Programs Offered by the Department:

SR.NO.	PROGRAM NAME	PROGRAM APPLIED LEVEL	YEAR OF START / YEAR OF CLOSED	SANCTIONED INTAKE	INCREASE/DECREASE INTAKE (if any)	YEAR OF INCREASE/DECREASE	CURRENT INTAKE	YEAR OF AICTE APPROVAL	AICTE/COMPETENT AUTHORITY ARROVAL DETAILS	ACCREDITATION STATUS	FROM	TO	NO. OF TIMES PROGRAM ACCREDITE
1	Biomedical Engineering	UG	2016 / --	60	No	NA	60	2016	F.No. Southern/1-46226565990/2026/EOA	Granted accreditation for 3 years for the period (specify period)	2023	2026	1

List of the Allied Departments/Cluster and Programs:

SR.NO.	ALLIED DEPARTMENT NAME	PROGRAM NAME	PROGRAM APPLIED LEVEL	YEAR OF START / YEAR OF CLOSED	SANCTIONED INTAKE	INCREASE/DECREASE INTAKE (if any)	YEAR OF INCREASE/DECREASE	CURRENT INTAKE	YEAR OF AICTE APPROVAL	AICTE/COMPETENT AUTHORITY ARROVAL DETAILS	ACCREDITATION STATUS	FROM
1	Electronics Engineering (VLSI Design and Technology)	Electronics Engineering (VLSI Design and Technology)	UG	2024 / --	60	No	NA	60	2024	F.No. Southern/1-46226565990/2026/EOA	Not eligible for accreditation	--
2	Electronics and Communication Engineering	VLSI Design	PG	2012 / --	18	No	NA	18	2012	F.No. Southern/1-46226565990/2026/EOA	Granted accreditation for 3 years for the period (specify period)	2024
3	Electronics and Communication Engineering	Electronics & Communication Engineering	UG	2006 / --	60	Yes	2025	180	2025	F.No. Southern/1-46226565990/2026/EOA	Granted accreditation for 3 years for the period (specify period)	2024

Sanctioned Intake for Last Five Years for the Electronics & Communication Engineering	
Academic Year	**Sanctioned Intake**
2025-26	180
2024-25	120
2023-24	120
2022-23	120
2021-22	120
2020-21	180

B2. Detail of Head of the Department for the program under consideration:

A. Name of the HoD :	RAJALAXMI S
B. Nature of appointment:	Regular
C. Qualification:	M.E. and Ph.D.

B3. Program Details

Table No.B3.1: Admission details for the program excluding those admitted through multiple entry and exit points.

Item (Information to be provided cumulatively for all the shifts with explicit headings, wherever applicable)	2025-26 (CAY)	2024-25 (CAYm1)	2023-24 (CAYm2)	2022-23 (CAYm3)	2021-22 (CAYm4)	2020-21 (CAYm5)	2019-20 (CAYm6)
N=Sanctioned intake of the program (as per AICTE /Competent authority)	60	60	60	60	60	60	60
N1=Total no. of students admitted in the 1st year minus the no. of students, who migrated to other programs/ institutions plus no. of students, who migrated to this program	42	52	61	43	34	30	48
N2=Number of students admitted in 2nd year in the same batch via lateral entry including leftover seats	0	0	0	1	0	0	0
N3=Separate division if any	0	0	0	0	0	0	0
N4=Total no. of students admitted in the 1st year via all supernumerary quotas	0	0	0	0	0	0	0
Total number of students admitted in the program (N1 + N2 + N3 + N4) - excluding those admitted through multiple entry and exit points.	42	52	61	44	34	30	48

CAY= Current Academic Year. CAYm1= Current Academic Year Minus 1 CAYm2= Current Academic Year Minus 2. LYG= Last Year Graduate. LYGm1= Last Year Graduate Minus 1. LYGm2= Last Year Graduate Minus 2.

B4. Enrolment Ratio in the First Year

Table No. B4.1: Student enrolment ratio in the 1st year.

Year of entry	N (From Table 4.1)	N1 (From Table 4.1)	N4 (From Table 4.1)	Enrollment Ratio [(N1/N)*100]
2025-26 (CAY)	60	42	0	70.00
2024-25 (CAYm1)	60	52	0	86.67
2023-24 (CAYm2)	60	61	0	101.67

Average [(ER1 + ER2 + ER3) / 3] = 86.11≡ 17.00

B5. Success Rate of the Students in the Stipulated Period of the Program

Table No.B5.1: The success rate in the stipulated period of a program.

Item	(2021-22) LYG	(2020-21) LYGm1	(2019-20) LYGm2
A*= (No. of students admitted in the 1st year of that batch and those actually admitted in the 2nd year via lateral entry, plus the number of students admitted through multiple entry (if any) and separate division if applicable, minus the number of students who exited through multiple entry (if any).	60.00	60.00	60.00
B=No. of students who graduated from the program in the stipulated course duration	29.00	30.00	46.00
Success Rate (SR)= (B/A) * 100	48.33	50.00	76.67

Average SR of three batches ((SR_1+ SR_2+ SR_3)/3): 58.33

B6. Academic Performance of the First-Year Students of the Program

Table No.B6.1: Academic Performance of the First-Year Students of the Program.

Academic Performance	CAYm1(2024-25)	CAYm2(2023-24)	CAYm3 (2022-23)
Mean of CGPA or mean percentage of all successful students(X)	7.33	7.45	6.66
Y=Total no. of successful students	52.00	56.00	43.00
Z=Total no. of students appeared in the examination	52.00	56.00	43.00
API [X*(Y/Z)]	7.33	7.45	6.66

Average API[(AP1+AP2+AP3)/3] : 7.15

B7: Academic Performance of the Second Year Students of the Program

Table No.B7.1: Academic Performance of the Second Year Students of the Program.

Academic Performance	CAYm1 (2024-25)	CAYm2 (2023-24)	CAYm3 (2022-23)
X=(Mean of 2nd year grade point average of all successful students on a 10-point scale) or (Mean of the percentage of marks of all successful students in 2rd year/10)	7.43	7.34	7.20
Y=Total no. of successful students	56.00	44.00	31.00
Z=Total no. of students appeared in the examination	61.00	44.00	34.00
API [X * (Y/Z)]	6.82	7.34	6.56

Average API [(AP1 + AP2 + AP3)/3] : 6.91

B8. Academic Performance of the Third Year Students of the Program

Table No.B8.1: Academic Performance of the Third Year Students of the Program

Academic Performance	CAYm1 (2024-25)	CAYm2 (2023-24)	CAYm3 (2022-23)
X=(Mean of 3rd year grade point average of all successful students on a 10-point scale) or (Mean of the percentage of marks of all successful students in 3rd year/10)	7.20	7.30	7.83
Y=Total no. of successful students	41.00	29.00	30.00
Z=Total no. of students appeared in the examination	44.00	31.00	30.00
API [X*(Y/Z)]:	6.71	6.83	7.83

Average API [(AP1 + AP2 + AP3)/3] : 7.12

B9. Placement, Higher Studies, and Entrepreneurship

Table No.B9.1: Placement, higher studies, and entrepreneurship details.

Item	LYG (2021-22)	LYGm1(2020-21)	LYGm2(2019-20)
FS*=Total no. of final year students	60.00	60.00	60.00
X=No. of students placed	27.00	25.00	40.00
Y=No. of students admitted to higher studies	2.00	5.00	5.00
Z= No. of students taking up entrepreneurship	0.00	0.00	1.00
Placement Index(P) = (((X + Y + Z)/FS) * 100):	48.33	50.00	76.67

Average Placement Index = (P_1 + P_2 + P_3)/3: 58.33 Placement Index Points:

PART C: Faculty Details in Department and Allied Departments

(Data to be filled in for the Department and Allied Departments)

C1. Faculty details of Department and Allied Departments

Table No.C1: Faculty details in the Department for the past 3 years including CAY

Sr.No	Name of the Faculty	PAN No.	Highest degree	University	Area of Specialization	Date of Joining in this Institution	Experience in years in current institute	Designation at Time Joining in this Institution	Present Designation	The date on which Designated as Professor/ Associate Professor if any	Nature of Association (Regular/ Contract/ Ad hoc)	Currently Associated (Y/N)	In case of NO, Date of Leaving	IS HOD?
1	RAJALAXMI S	XXXXXXX71L	M.E. and Ph.D.	Anna University	Image Processing	12/06/2023	3.1	Professor	Professor	12/06/2023	Regular	Yes		Yes
2	PAVITHRA S	XXXXXXX56H	M.E.	Anna University	Biomedical Engineering	11/06/2018	8.1	Assistant Professor	Assistant Professor		Regular	Yes		No
3	PRIYAVARTHINI S	XXXXXXX06N	M.Tech	Anna University	Biotechnology	07/07/2025	1	Assistant Professor	Assistant Professor		Regular	Yes		No
4	LEKSHMI V NAIR	XXXXXXX01H	M.E.	Noorul Islam University	Biomedical Instrumentation	21/07/2023	2.11	Assistant Professor	Assistant Professor		Regular	Yes		No
5	MANOJ PRABU M	XXXXXXX18L	M.E. and Ph.D.	Anna University	Embedded System Technologies	10/05/2024	2.2	Associate Professor	Associate Professor	10/05/2024	Regular	Yes		No
6	DOOSLIN MERCY BAI V	XXXXXXX13Q	M.Sc. and PhD	Manonmaniam Sundaranar University	Microbiology	10/04/2019	6	Professor	Professor	10/04/2019	Regular	No	30/04/2025	No
7	SANKARA GOMATHY B	XXXXXXX50J	M.Tech and Ph.D.	Manonmaniam Sundaranar University	Instrumentation and Control System	10/05/2024	1.2	Professor	Professor	10/05/2024	Regular	No	31/07/2025	No
8	VISHALI A M	XXXXXXX29P	M.Tech	Anna University	Nanoscience and Technology	18/07/2018	7.11	Assistant Professor	Assistant Professor		Regular	Yes		No
9	JASON RAJ J	XXXXXXX69H	M.Tech	Karunya University	Biomedical Instrumentation	10/07/2024	2	Assistant Professor	Assistant Professor		Regular	Yes		No
10	BALAJI M	XXXXXXX81Q	M.E.	Vinayaka Missions University	Applied Electronics	18/07/2018	7.11	Assistant Professor	Assistant Professor		Regular	Yes		No
11	DIVYA A	XXXXXXX99J	M.E.	Anna University	Applied Electronics	12/11/2018	7.8	Assistant Professor	Assistant Professor		Regular	Yes		No
12	ABIRAMEE R	XXXXXXX36C	M.Tech	Sastra University	Medical Nanotechnology	19/08/2019	6.10	Assistant Professor	Assistant Professor		Regular	Yes		No
13	SOWKAARTHIGA A	XXXXXXX69G	M.E.	Anna University	VLSI Design	12/11/2018	7.8	Assistant Professor	Assistant Professor		Regular	Yes		No
14	SHOWBANA R	XXXXXXX47D	M.Tech	Amrita University	Biomedical Engineering	03/06/2019	6.10	Assistant Professor	Assistant Professor		Regular	No	30/04/2026	No
15	DIVYA JENIFAR P	XXXXXXX72A	M.E.	Avinashi Lingam University	Medical Electronics	30/05/2022	3.11	Assistant Professor	Assistant Professor		Regular	No	30/04/2026	No

16	KEERTHANA R	XXXXXXX18P	M.E.	Avinashi Lingam University	Medical Electronics	30/05/2022	1.3	Assistant Professor	Assistant Professor		Regular	No	31/08/2023	No
17	KEVIN PAUL J A	XXXXXXX45D	M.Tech	Vellore Institute of Technology	Biomedical Engineering	16/06/2023	2.10	Assistant Professor	Assistant Professor		Regular	No	30/04/2026	No
18	SIVA SAKTHI B	XXXXXXX49J	M.E.	Anna University	Medical Electronics	11/05/2026	0.2	Assistant Professor	Assistant Professor		Regular	Yes		No
19	SHALINI R V	XXXXXXX95L	M.E. and Ph.D.	Anna University	Applied Electronics	08/06/2017	7	Assistant Professor	Associate Professor	01/06/2022	Regular	No	27/06/2024	No
20	ARUNA M	XXXXXXX65E	M.E.	Avinashi Lingam University	Medical Electronics	19/05/2023	0.11	Assistant Professor	Assistant Professor		Regular	No	30/04/2024	No
21	SINDHUJA R	XXXXXXX29N	M.Tech	Karunya University	Biomedical Instrumentation	16/08/2021	2.11	Assistant Professor	Assistant Professor		Regular	No	31/07/2024	No
22	ANITHA S	XXXXXXX14L	M.Tech	Karunya University	Biomedical Instrumentation	16/08/2021	2.10	Assistant Professor	Assistant Professor		Regular	No	18/06/2024	No

Table No.C2: Faculty details of Allied Departments for the past 3 years including CAY.

Sr.No	Name of the Faculty	PAN No.	APAAR faculty ID*(if any)	Highest degree	University	Area of Specialization	Date of Joining in this Institution	Experience in years in current institute	Designation at Time Joining in this Institution	Present Designation	The date on which Designated as Professor/ Associate Professor if any	Nature of Association (Regular/ Contract/ Ad hoc)	Currently Associated (Y/N)	In case of NO, Date of Leaving	IS HOD?
1	Dr.S Bhavani	XXXXXXX27E	NA	M.E. and Ph.D.	Anna University	Information and Communication Engineering	01/06/2019	7.1	Professor	Professor	01/06/2019	Regular	Yes		Yes
2	Dr.R.Ranjith Kumar	XXXXXXX36A	NA	M.E. and Ph.D.	Anna University	Electrical and Electronics Engineering Science	04/07/2018	7.9	Associate Professor	Professor	03/08/2022	Regular	No	30/04/2026	No
3	Dr Bindhu Salim	XXXXXXX38N	NA	M.E. and Ph.D.	Anna University	Information and Communication Engineering	03/05/2023	2.6	Professor	Professor	03/05/2023	Regular	No	19/11/2025	No
4	Dr R Jamuna	XXXXXXX51E	NA	M.E. and Ph.D.	Anna University	Information and Communication Engineering	03/01/2022	2.4	Professor	Professor	03/01/2022	Regular	No	10/05/2024	No
5	Dr.B.Sankaragomathy	XXXXXXX50J	NA	M.E. and Ph.D.	Anna University	Instrumentation and Control system	15/06/2020	3.10	Professor	Professor	15/06/2020	Regular	No	10/05/2024	No

6	Dr Senthil Kumar R	XXXXXXX23R	NA	M.E. and Ph.D.	Anna University	Embedded System Technologies	04/04/2022	3.1	Associate Professor	Associate Professor	04/04/2022	Regular	No	30/05/2025	No
7	Dr Remya.K.R	XXXXXXX61N	NA	M.E. and Ph.D.	Calicut University	Electronics and Communication engineering	22/06/2022	4	Associate Professor	Associate Professor	22/06/2022	Regular	Yes		No
8	Dr. Tamilarasi	XXXXXXX15B	NA	M.E. and Ph.D.	Anna University	Information and communication Engineering	17/08/2022	3.8	Associate Professor	Associate Professor	17/08/2022	Regular	No	30/04/2026	No
9	Dr.M.Manojprabhu	XXXXXXX18L	NA	M.E. and Ph.D.	Anna University	Embedded system Technologies	30/06/2021	2.10	Associate Professor	Associate Professor	30/06/2021	Regular	No	10/05/2024	No
10	Dr Grace Jency	XXXXXXX68J	NA	M.E. and Ph.D.	Anna University	Electronics and Communication Engineering	10/06/2022	3.11	Associate Professor	Associate Professor	10/06/2022	Regular	No	30/05/2026	No
11	Dr Senthil Kumar R		NA	M.E. and Ph.D.	Anna University	Embedded system Technologies	04/04/2022	1.8	Associate Professor	Associate Professor	04/04/2022	Regular	No	29/12/2023	No
12	Dr R.Sudha	XXXXXXX79D	NA	M.E. and Ph.D.		Communication Systems	08/05/2008	18.2	Assistant Professor	Associate Professor	18/05/2017	Regular	Yes		No
13	Dr Sribhuvaneshwari H	XXXXXXX27N	NA	M.E. and Ph.D.	Anna University	Embedded System T echnologies	02/03/2023	2.2	Assistant Professor	Assistant Professor		Regular	No	30/05/2025	No
14	Dr Manokaran J	XXXXXXX91Q	NA	M.E. and Ph.D.	SRM University	Applied Electronics	22/07/2024	1.9	Assistant Professor	Assistant Professor		Regular	No	30/04/2026	No
15	Mr.S.Siva kumar	XXXXXXX96Q	NA	M.E.	Anna University	Communication Systems	30/05/2013	12.11	Assistant Professor	Assistant Professor		Regular	No	30/04/2026	No
16	Mr.T.Maheswaran	XXXXXXX72P	NA	M.E.	Anna University	Embedded system Technologies	03/06/2013	13.1	Assistant Professor	Assistant Professor		Regular	Yes		No
17	Ms.Renita Pearlin T	XXXXXXX49A	NA	M.E.	Anna University	VLSI Design	05/08/2020	5.11	Assistant Professor	Assistant Professor		Regular	Yes		No
18	Ms. Ashna A	XXXXXXX14J	NA	M.E.	Anna University	VLSI Design	05/06/2014	11.7	Assistant Professor	Assistant Professor		Regular	No	02/01/2026	No
19	Mr.K.Sivaprasath	XXXXXXX80J	NA	M.E.			20/06/2015	10.11	Assistant Professor	Assistant Professor		Regular	No	29/05/2026	No
20	Ms. S. Yogitha		NA	M.E.	Anna University	Communication Systems	20/06/2015	9.5	Assistant Professor	Assistant Professor		Regular	No	28/11/2024	No
21	Ms.K.B.Sangavi	XXXXXXX16N	NA	M.E.	Anna University	Communication Systems	04/07/2018	8	Assistant Professor	Assistant Professor		Regular	Yes		No
22	Mr Mohammed Yas Nivisan	XXXXXXX99E	NA	M.E.	Anna University	Embedded system Technologies	03/06/2019	6.10	Assistant Professor	Assistant Professor		Regular	No	30/04/2026	No

23	Mrs Pushpakumari.R	XXXXXXX31N	NA	M.E.	Anna University	Communication Systems	26/05/2021	2.11	Assistant Professor	Assistant Professor		Regular	No	30/04/2024	No
24	Mr.S.Navaneethan	XXXXXXX34J	NA	M.E.	Anna University	VLSI Design	05/06/2014	11.5	Assistant Professor	Assistant Professor		Regular	No	28/11/2025	No
25	Mr Santhoshkumar.A	XXXXXXX96M	NA	M.E.	Anna University	VLSI Design	07/08/2020	4.2	Assistant Professor	Assistant Professor		Regular	No	18/10/2024	No
26	Mr Saravanan S	XXXXXXX06C	NA	M.E.	Anna University	VLSI Design	08/08/2020	5.8	Assistant Professor	Assistant Professor		Regular	No	30/04/2026	No
27	Ms J.Varshini	XXXXXXX83D	NA	M.E.	Anna University	VLSI Design	21/06/2021	3.6	Assistant Professor	Assistant Professor		Regular	No	30/12/2024	No
28	Ms.M.Madhuvarshini	XXXXXXX50E	NA	M.E.	Anna University	VLSI Design	21/06/2021	4.7	Assistant Professor	Assistant Professor		Regular	No	30/01/2026	No
29	Ms L.Prasanna Maria	XXXXXXX47Q	NA	M.E.	Anna University	VLSI Design	21/06/2021	5	Assistant Professor	Assistant Professor		Regular	Yes		No
30	Mrs Sridevi K	XXXXXXX83M	NA	M.E.	Anna University	VLSI Design	05/01/2022	4.6	Assistant Professor	Assistant Professor		Regular	Yes		No
31	Mrs. Subhashini U	XXXXXXX93L	NA	M.E.	Anna University	VLSI Design	05/01/2022	3.11	Assistant Professor	Assistant Professor		Regular	No	31/12/2025	No
32	Ms. R.Deeksha		NA	M.E.	Anna University	Embedded system Technologies	05/01/2022	2.9	Assistant Professor	Assistant Professor		Regular	No	18/10/2024	No
33	Mrs.Prema C	XXXXXXX80K	NA	M.E.	Anna University	VLSI Design	05/01/2022	4.6	Assistant Professor	Assistant Professor		Regular	Yes		No
34	Mrs Mahima Mohan	XXXXXXX76C	NA	M.E.	Anna University	V	06/06/2023	1.1	Assistant Professor	Assistant Professor		Regular	No	29/07/2024	No
35	Ms.Ezhilarasi V	XXXXXXX22A	NA	M.E.	Anna University	Applied Electronics	21/06/2021	3.3	Assistant Professor	Assistant Professor		Regular	No	18/10/2024	No
36	Ms S Jeevitha	XXXXXXX39C	NA	M.E.	Anna University	Embedded system Technologies	03/06/2017	9.1	Assistant Professor	Assistant Professor		Regular	Yes		No
37	Dr. C Aravind	XXXXXXX70A	NA	MS and PhD	Anna University	Information and Communication Engineering	14/08/2024	0.8	Professor	Professor		Regular	No	30/04/2025	No
38	Mrs. Pricilla Sofia	XXXXXXX78B	NA	M.E.	Karunya University	VLSI Design	05/03/2024	2.4	Assistant Professor	Assistant Professor		Regular	Yes		No
39	Mrs Heamalatha S	XXXXXXX25P	NA	M.E.	Anna University	Embedded system Technologies	22/07/2024	1.11	Assistant Professor	Assistant Professor		Regular	Yes		No
40	Mrs. Brinda P	XXXXXXX25D	NA	M.E.	Anna University	Power Electronics and Drives	02/01/2026	0.6	Assistant Professor	Assistant Professor		Regular	Yes		No
41	Mrs. Ramalakshmi R	XXXXXXX69B	NA	M.E.	Anna University	Embedded system Technologies	02/06/2015	11.1	Assistant Professor	Assistant Professor		Regular	Yes		No

42	Mr. Boopathy S	XXXXXXX53K	NA	M.E.	Anna University	Communication Systems	21/03/2025	1.3	Assistant Professor	Assistant Professor		Regular	Yes		No
43	Mrs. Eswari P	XXXXXXX85L	NA	M.E.	Anna University	Embedded system Technologies	27/04/2026	0.2	Assistant Professor	Assistant Professor		Regular	Yes		No
44	Mr. Mahendar R	XXXXXXX71C	NA	M.E.	Anna University	VLSI Design	05/01/2026	0.6	Assistant Professor	Assistant Professor		Regular	Yes		No
45	Mr Arun L	XXXXXXX86B	NA	M.E.	Anna University	VLSI Design	07/01/2026	0.6	Assistant Professor	Assistant Professor		Regular	Yes		No
46	Mrs Padmapriya Kiruthika	XXXXXXX45H	NA	M.E.	Karpagam University	Power Electronics and Drives	08/04/2026	0.3	Assistant Professor	Assistant Professor		Regular	Yes		No
47	Mrs Parimaladevi K	XXXXXXX99P	NA	M.E.	Anna University	VLSI Design	08/04/2026	0.3	Assistant Professor	Assistant Professor		Regular	Yes		No
48	Ms Thaarani C	XXXXXXX35J	NA	M.E.	Anna University	VLSI Design	08/04/2026	0.3	Assistant Professor	Assistant Professor		Regular	Yes		No
49	Mr Thirumurugan S	XXXXXXX55K	NA	M.E.	Anna University	Embedded system Technologies	03/01/2024	2.6	Assistant Professor	Assistant Professor		Regular	Yes		No
50	Mr Pravin TV	XXXXXXX28L	NA	M.E.	Anna University	VLSI Design	07/01/2026	0.6	Assistant Professor	Assistant Professor		Regular	Yes		No
51	Mr Akarsh S Kumar	XXXXXXX31C	NA	M.E.	Anna University	VLSI Design	07/01/2026	0.6	Assistant Professor	Assistant Professor		Regular	Yes		No
52	Mr Prabhu G	XXXXXXX40A	NA	M.E.	Anna University	VLSI Design	07/01/2026	0.6	Assistant Professor	Assistant Professor		Regular	Yes		No
53	Mrs Renukadevi A	XXXXXXX14R	NA	M.E.	Anna University	Communication Systems	08/04/2026	0.3	Assistant Professor	Assistant Professor		Regular	Yes		No
54	Mr Mohamed Naufal M	XXXXXXX64B	NA	M.E.	Anna University	VLSI Design	07/01/2026	0.6	Assistant Professor	Assistant Professor		Regular	Yes		No
55	Mrs. Vasanthi R	XXXXXXX62B	NA	M.E.	Anna University	VLSI Design	08/04/2026	0.3	Assistant Professor	Assistant Professor		Regular	Yes		No
56	Dr.P.Dhilipkumar	XXXXXXX92F	NA	M.E. and Ph.D.	Anna University	VLSI Design	19/05/2025	1.1	Associate Professor	Associate Professor		Regular	Yes		No
57	Mr Karthikeyan Vairavel	XXXXXXX43F	NA	M.E.	Anna University	Communication Systems	07/01/2026	0.6	Assistant Professor	Assistant Professor		Regular	Yes		No
58	Raji D	XXXXXXX60M	NA	M.E.	Anna University	VLSI Design	02/06/2025	1	Assistant Professor	Assistant Professor		Regular	No	05/06/2026	No
59	MRS. SHARMILA S	XXXXXXX56G	NA	M.E.	Anna University	VLSI Design	02/01/2025	1.3	Assistant Professor	Assistant Professor		Regular	No	30/04/2026	No

C2. Student-Faculty Ratio (SFR)

No. of UG(Engineering) programs in Department including allied departments/ clusters (UGn):
UG1=1st UG program
UGn=nth UG program
B= No. of Students in UG 2nd year (ST)
C= No. of Students in UG 3rd year (ST)
D= No. of Students in UG 4th year (ST)
No. of PG (Engineering) programs in Department including allied departments/ clusters (PGm):
PG1=1st PG program.
PGm=mth PG program
A= No. of Students in PG 1st year
B= No. of Students in PG 2nd year
Student Faculty Ratio **(SFR)** = S/F
S= No. of students of all programs in the Department including all students of allied departments/clusters.
No. of students (ST)=Sanctioned Intake (SA)+ Actual admitted students via lateral entry including leftover seats (L) if any (limited to 10 % of SA)
Students who admitted under supernumerary quotas (SNQ, EWS, etc) will not be considered in calculating SFR value. Those students are exempted.
F=Total no. of regular or contractual faculty members (Full Time) in the Department, including allied departments/clusters (excluding first year faculty (The faculty members who have a 100% teaching load in the first-year courses)).

No. of UG Programs in the Department1 No. of PG Programs in the Department1

Table No.C2.1: Student-faculty ratio.

Description	**CAY(2025-26)**	**CAYm1 (2024-25)**	**CAYm2 (2023-24)**
UG1.B	60	60	61
UG1.C	60	61	60
UG1.D	61	60	60
UG1: Biomedical Engineering	**181**	**181**	**181**
UG2.B	132	126	126
UG2.C	126	126	123
UG2.D	126	123	186
UG2: Electronics & Communication Engineering	**384**	**375**	**435**
UG3.B	62	0	0
UG3.C	0	0	0
UG3.D	0	0	0
UG3: Electronics Engineering (VLSI Design and Technology)	**62**	**0**	**0**
DS=Total no. of students in all UG and PG programs in the Department	181	181	181
AS=Total no. of students of all UG and PG programs in allied departments	482	411	471
S=Total no. of students in the Department (DS) and allied departments (AS)	**S1=** 663	**S2=** 592	**S3=** 652
DF=Total no. of faculty members in the Department	14	14	15
AF= Total no. of faculty members in the allied Departments	26	30	35
F=Total no. of faculty members in the Department (DF) and allied Departments (AF)	**F1=** 40	**F2=** 44	**F3=** 50
FF=The faculty members in F who have a 100% teaching load in the first-year courses	0	0	0
Student Faculty Ratio (SFR)=S/(F-FF)	**SFR1= 16.57**	**SFR2= 13.45**	**SFR3= 13.04**
Average SFR for 3 years	**SFR= 14.35**		

C3. Faculty Qualification

- Faculty qualification index (FQI) = 2.5 * [(10X +4Y)/RF] where
- X=No. of faculty members with Ph.D. degree or equivalent as per AICTE/UGC norms.
- Y=No. of faculty members with M. Tech. or ME degree or equivalent as per AICTE/ UGC norms.
- RF=No. of required faculty in the Department including allied Departments to adhere to the 20:1 Student-Faculty ratio, with calculations based on both student numbers and faculty requirements as per section C2 of this documents: (RF=S/20).

Table No.C3.1: Faculty qualification.

Year	X	Y	RF	FQ = 2.5 x [(10X + 4Y) / RF)]
2025-26(CAY)	10	30	33.00	16.67
2024-25(CAYm1)	14	31	29.00	22.76
2023-24(CAYm2)	14	37	32.00	22.50

C4. Faculty Cadre Proportion

- Faculty Cadre Proportion is 1(RF1): 2(RF2): 6(RF3)
- RF1= No. of Professors required = 1/9 * No. of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (S) as per C2 of this documents:.
- RF2= No. of Associate Professors required = 2/9 * No. of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (S) as per section C2 of this documents:.
- RF3= No. of Assistant Professors required = 6/9 * No. of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (S) as per section C2 of this documents:.
- Faculty cadre and qualification and experience should be as per AICTE/UGC norms.

Table No.C4.1: Faculty cadre proportion details.

Year	Professors		Associate Professors		Assistant Professors	
	Required RF1	Available AF1	Required RF2	Available AF1	Required RF3	Available AF3
2025-26	4.00	3.00	7.00	6.00	22.00	31.00
2024-25	3.00	6.00	7.00	5.00	20.00	33.00
2023-24	4.00	6.00	7.00	6.00	22.00	38.00
Average	RF1=3.67	AF1=5.00	RF2=7.00	AF2=5.67	RF2=21.33	AF2=34.00

C5. Visiting/Adjunct Faculty/Professor of Practice

Table No. C5.1: List of visiting/adjunct faculty/professor of practice and their teaching and practical loads.

(CAYm1)

S.No	Name of the Person	Designation	Organization	Name of the Course	No. of hours handled
1	Dr.S.Atheena Milagi Pandian	Founder & Global Director	Atheenapandian Private Limited, Chennai, Coimbatore, Courtallam	21BM312 Electron Devices and Biocircuits Laboratory	36.00
2	Dr.S.Atheena Milagi Pandian	Founder & Global Director	Atheenapandian Private Limited, Chennai, Coimbatore, Courtallam	21BM613 Advanced Biomedical Equipment Laboratory	36.00

(CAYm2)

S.No	Name of the Person	Designation	Organization	Name of the Course	No. of hours handled
1	Dr.S.Atheena Milagi Pandian	Founder & Global Director	Atheenapandian Private Limited, Chennai, Coimbatore, Courtallam	21BM312 Electron Devices and Biocircuits Laboratory	36.00
2	Dr.S.Atheena Milagi Pandian	Founder & Global Director	Atheenapandian Private Limited, Chennai, Coimbatore, Courtallam	21BM613 Advanced Biomedical Equipment Laboratory	36.00

(CAYm3)

S.No	Name of the Person	Designation	Organization	Name of the Course	No. of hours handled
1	Dr.S.Atheena Milagi Pandian	Founder & Global Director	Atheenapandian Private Limited, Chennai, Coimbatore, Courtallam	21BM312 Electron Devices and Biocircuits Laboratory	36.00
2	Dr.S.Atheena Milagi Pandian	Founder & Global Director	Atheenapandian Private Limited, Chennai, Coimbatore, Courtallam	21BM613 Advanced Biomedical Equipment Laboratory	36.00

C6. Academic Research

Table No. C6.1: Faculty publication details.

S.No.	Item	2024-25 (CAYm1)	2023-24 (CAYm2)	2022-23 (CAYm3)
1	No. of peer reviewed journal papers published	77	15	12
2	No. of peer reviewed conference papers published	7	6	4
3	No. of books/book chapters published	2	1	1

C7. Sponsored Research Project

Table No. C7.1: List of sponsored research projects received from external agencies.

(CAYm1)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Dr.B.Sankara Gomathi	Mr.M.Balaji	Biomedical Engineering	Design and Development of an Affordable Assistive Device for Elderly Mobility	Lakshmi Machine Works	12 months	1.00
Ms.P.Divya Jenifar	Shamili M, Manimaran P, Sowbarnicka M, Kavisri S	Biomedical Engineering	Ergonomic Wheelchair-Stretcher for Enhanced Patient Mobility	Tamil Nadu State Council of Science and Technology	6 months	0.08
						Amount received (Rs.):1.08

(CAYm2)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Dr.V.Dooslin Mercy Bai	Ms.P.Divya Jenifar	Biomedical Engineering	IoT-Based Remote Patient Monitoring System for Elderly and High-Risk Patients	Roots industries Private Limited	12 months	3.00
Dr.S.Rajalaxmi	Ms.S.Pavithra	Biomedical Engineering	AI-Based Early Detection of Medical Conditions Using Biomedical Signals	ELGi Equipment	6 months	1.00
						Amount received (Rs.):4.00

(CAYm3)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Dr.V.Dooslin Mercy Bai	Nithin.B	Biomedical Engineering	Advanced Endoscopy Capsule	IIT Kanpur	12 months	4.00
Dr.R.V.Shalini	Mrs.A.M.Vishali	Biomedical Engineering	Development of an Energy-Efficient Oxygen Concentrator for Home-Based Patient Care	Faraday Ozone	18 months	3.00
						Amount received (Rs.):7.00

Total Amount (Lacs) Received for the Past 3 Years: 12.08

Note*:

- Only sponsored research projects will be considered. Infrastructure-based projects will not be considered here.

C8. Consultancy Work

Table No. C8.1: List of consultancy projects received from external agencies.

(CAYm1)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Dr.S.Rajalaxmi	B.Priyadharshini,M.Raguram,G.Sangamithra,A.S.Shamna	Biomedical Engineering	Design of Robotic X-Ray using AI	Atheenapandiyan Private Limited	1 Year	4.50
Ms.S.Pavithra	V.Boomika,S.Jeya sree bala,G.Prisha	Biomedical Engineering	Anti Sleep Alarm Device for Drivers	Prashan Medical Technologies,	1 Year	3.00
Dr.B.Sankara Gomathi	R.Rithika,S.Sabarish,K.Subbulakshmi	Biomedical Engineering	Respiratory Exertion Using Virtual Reality	Prodos Instrumentation Private Limited	6 months	0.50
						Amount received (Rs.):8.00

(CAYm2)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Dr.V. Dooslin Mercy Bai	Kowshik Saravana,Sabareesh	Biomedical Engineering	Evaluation and Quantitative Analysis of Bioactive compounts from Haetocerous Calcitrans Against Human Patheogen's	Orbito Asia Diagnostics	1 Year	5.00
Ms.P.Divya Jenifar	Shamili M, Manimaran P, Sowbarnicka M, Kavisri S	Biomedical Engineering	Ergonomic Wheelchair-Stretcher for Enhanced Patient Mobility	Prashan Medical Technologies, Coimbatore	1 Year	2.00
Dr.S.Rajalxmi	Mrs.R.Abiramee	Biomedical Engineering	Development of Wearable Health Monitoring Device	Randall Medical Technologies	1 Year	2.00
						Amount received (Rs.):9.00

(CAYm3)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Dr.R.V.Shalini	Sathya priya,Manju,Akshaya	Biomedical Engineering	IOT Based Mobile Patient Monitoring Systems	Atheenapandiyan Private Limited	1 Year	3.50
						Amount received (Rs.):3.50

Total amount (Lacs) received for the past 3 years: 20.50

Note*:

- Only consultancy projects will be considered. Infrastructure-based projects will not be considered here.

C9. Institution Seed Money or Internal Research Grant to its Faculty for Research Work

Table No. C9.1: List of faculty members received seed money or internal research grant from the Institution.

(CAYm1)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25	Outcomes of the project
Dr.B.Sankara Gomathi	Biosensing and Regenerating Wound Healing Patch	1 Year	0.65	0.65	Smart Regenerative Wound Monitoring Patch – prototype developed
Dr.S.Rajalaxmi	Heatable Neck Brace for Cervical Muscle Relaxation	3 Months	0.25	0.25	Heatable Neck Brace with Thermal Stimulation – prototype developed
Dr.M.Manoj Prabu	Eco-Friendly Banana Fiber Manufacturing Technology	5 Months	0.45	0.45	Eco-friendly Banana Fiber Manufacturing Process/Prototype developed
			Amount received (Rs.): 1.35		

(CAYm2)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25	Outcomes of the project
Dr.V.Dooslin Mercy Bai	Smart Algal Biorefinary Ecosystem for Multimodel Bioenergy Generation	1 Year	1.00	1.00	Multimodal Bioenergy Generation System – prototype developed
Dr.S.Rajalaxmi	Eye Blink Communication and heart rate Monitoring for Paralysed Patients	6 Months	0.50	0.50	Eye-Blink-Based Communication and Heart-Rate Monitoring System – prototype developed
Ms.S.Pavithra	Green Synthesis and Characterization of Bioactive Nanoparticles Using Plant Extracts	2 Months	0.50	0.50	Bioactive nanoparticles synthesized, characterized and evaluated for antimicrobial activity
Mr.M.Balaji	Anytime Portable Pharmacy	3 Months	0.40	0.40	Prototype Developed
			Amount received (Rs.): 2.40		

(CAYm3)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25	Outcomes of the project
Dr.V.Dooslin Mercy Bai	Evaluation of Bioactive Compounds from Chaetoceros calcitrans Against Human Pathogens	12 Months	2.00	2.00	Antimicrobial bioactive compound/extract from Chaetoceros calcitrans
Ms.S.Pavithra	Development and Characterization of Nanoparticle-Based Antimicrobial Wound Dressing	6 Months	0.50	0.50	Antimicrobial Compounds Identified
			Amount received (Rs.): 2.50		

Total amount (Lacs) received for the past 3 years : 6.25

PART D: Laboratory Infrastructure in the Department

(Data to be filled in for the Department)

D1. Adequate and Well-Equipped Laboratories, and Technical Manpower

Table No.D1.1: List of laboratories and technical manpower.

Sr. No	Name of the Laboratory	Number of students per set up(Batch Size)	Name of the Important Equipment	Weekly utilization status(all the courses for which the lab is utilized)	Technical Manpower Support		
					Name of the Technical staff	Designation	Qualification
1	Biomedical Instrumentation Laboratory	60	Conductivity meter, Ph Meter, Patient Monitoring System, Glucanometer, Sphygmomanometer,	21BM321 - 3 H	Ms.Nabha Sindhu	Lab Instructor	B.E Biomedical Enginee
2	Pathology and Microbiology Laboratory	30	UV Spectrophotometer, Waterbath, Electronic Weighing Balance, Optical Microscope, Centrifuge,	21BM212 - 3 H	Ms.Shalini K	Lab Instructor	B.E Biomedical Enginee
3	Advanced Biomedical Equipment Laboratory	30	Defibrillator, Ventilator, Electroenceplalogram, Ultrasound Machine, Hemo Dialysis Machine	21BM613 - 3 H	Ms.Nabha Sindhu	Lab Instructor	B.E Biomedical Enginee

4	Bio Signal Processing/ Image Processing (Sharing)	60	PCs, DSP Processors (Kit/Add-on cards)	21BM412 - 3 H	Ms.Janaranjani	Lab Instructor	Diploma in Electronics a
5	Sensors and Transducers Laboratory (Sharing)	60	Kelvin's Double Bridge, Wheatstone Bridge Accessories, Decade Resistance Box,	21BM322 - 3 H	Mr.Dhandapani	Lab Instructor	Diploma in Electrical an
6	Microprocessor and Microcontroller laboratory (Sharing)	60	8086 Microprocessor, 8051 microcontroller, Keyboard and display, Traffic light control, Digital	21BM512 - 3 H	Ms.Janaranjani	Lab Instructor	Diploma in Electronics a
7	Electronics Laboratory (Sharing)	30	BC 107, BC 148, 2N2646, BFW10, IN4007, Zener Diodes, Resistors, Capacitors, Inductors Bread	21BM312 - 3 H	Mr.Arun	Lab Instructor	B.E. Electronics and Co
8	Integrated Circuits Laboratory (Sharing)	30	Cathode Ray Oscilloscope, Digital Storage Oscilloscope, Analog Trainer Kit, Integrated	21BM413 - 3 H	Mr.Arun	Lab Instructor	B.E. Electronics and Co

D2. Safety Measures in Laboratories

Table No. D2.1: List of various safety measures in laboratories.

Sr. No	Laboratory Name	Safety Measures
1	Biomedical Instrumentation Laboratory	MCB/ELCB protection; proper electrical earthing; fire extinguisher; first-aid kit; safety instruction board; inspection of cables and connectors; safe operation of biomedical instruments; faculty/technical staff supervision; proper equipment shutdown after experiments.
2	Pathology and Microbiology Laboratory	First-aid kit; fire extinguisher; handwash/eyewash facility; gloves, masks and PPE; safe handling of specimens and cultures; sterilization/disinfection practices; biomedical waste segregation and disposal; safe handling of sharps; supervised laboratory activities.
3	Advanced Biomedical Equipment Laboratory	MCB/ELCB protection; proper earthing; fire extinguisher; first-aid kit; equipment-specific safety instructions; supervised operation of specialized medical equipment; electrical isolation before maintenance; safe handling and shutdown of equipment.
4	Biosignal Processing / Image Processing Laboratory	MCB/ELCB protection; proper earthing; fire extinguisher; first-aid kit; safe operation of computers, DSP kits and signal acquisition systems; proper cable management; safe handling of electrical connections; equipment shutdown after use.
5	Sensors and Transducers Laboratory	MCB/ELCB protection; proper earthing; fire extinguisher; first-aid kit; safety instructions; inspection of wires and terminals before energizing circuits; safe handling of sensors and electronic trainer kits; supervised experimentation; proper shutdown of equipment.
6	Microprocessor and Microcontroller Laboratory	MCB/ELCB protection; proper earthing; fire extinguisher; first-aid kit; safe power-supply connections; inspection of trainer kits and interface modules; protection against short circuits; safe handling of electronic components; supervised operation and shutdown.

7	Electronics Laboratory	MCB/ELCB protection; proper earthing; fire extinguisher; first-aid kit; safety instruction board; checking circuit connections before power-on; safe handling of components and power supplies; proper operation of CRO/function generator; damaged cables not used.
8	Integrated Circuits Laboratory	MCB/ELCB protection; proper earthing; fire extinguisher; first-aid kit; safe operation of CRO/DSO and function generators; correct IC orientation and circuit verification before power-on; power switched off before circuit modification; safe component handling.

D3. Project Laboratory/Research Laboratory

7.5 Research Laboratories / Facilities

The Department of Biomedical Engineering has established specialized research facilities to support student projects, faculty research, interdisciplinary activities and application-oriented innovation. The facilities provide opportunities for students to move beyond routine curriculum-based laboratory experiments and gain hands-on exposure to emerging technologies, experimental methods and problem-oriented research.

The department currently utilizes the following specialized research facilities:

1. Algae Biomass Culture Laboratory – supporting research and student projects in Spirulina biomass cultivation, processing, biorefinery and value-added applications.
2. Industry-Supported Biomedical Research Laboratory – supported by Prashan Medical Technologies, providing specialized biomedical instrumentation for gait analysis, Internet of Medical Things (IoMT), electromyography (EMG) and electrooculography (EOG) based research and student projects.

These facilities provide a platform for experiential learning, interdisciplinary research, product development and industry-oriented project work, enabling students and faculty to apply biomedical engineering knowledge to healthcare, environmental, agricultural and sustainable technology applications.

7.5.1 ALGAE BIOMASS CULTURE LABORATORY

Research Focus: Spirulina Biomass Cultivation and Biorefinery

The Algae Biomass Culture Laboratory has been established to support research and student project activities related to algae biomass cultivation, processing and biorefinery. The laboratory provides facilities for the cultivation and harvesting of Spirulina biomass, which is further utilized for exploring value-added applications. The research activities focus on developing sustainable products and processes through the biorefinery approach, including biomass processing, extraction and conversion into useful products. The laboratory also supports interdisciplinary student projects and research aimed at applications in healthcare, agriculture, environmental management and renewable energy.

The laboratory facilitates the progression from Spirulina cultivation and biomass generation to processing, product development and application-oriented research, thereby providing students with hands-on exposure to sustainable biotechnology and product development.

Figure 7.5.1 Algae Biomass Culture Laboratory and Outcomes

The Algae Biomass Culture Laboratory provides a platform for integrated research on Spirulina cultivation, biomass generation, processing and biorefinery. The facility supports the development of value-added applications in healthcare, agriculture, environmental management and renewable energy through student projects and research activities.

Figure. 7.5.2 Process Flow Chart

7.5.1.1 Research Projects and Value-Added Outcomes – Algae Biomass Culture Laboratory

S.No.	Project / Research Title	Research Focus / Application	Outcome / Output	Relevant PO/PSO	Relevant SDG	Evidence
1	Spirulina Based Herbal Oral Rinse: A Natural Approach to Mucosal Wound Healing	Development and evaluation of a Spirulina-based herbal oral formulation for mucosal wound-healing applications	**Herbal oral rinse formulation** for oral care and mucosal wound-healing studies	PO1, PO2, PO3, PO4, PO5; PSO1	SDG 3 – Good Health and Well-being	

2	Dual Herbal Soap Formulations Spirulina With Tulsi-Pig Lard And Neem-Moringa Combinations For Enhanced Antifungal	Development of a value-added personal-care product incorporating Spirulina biomass/extracts	Spirulina-based herbal soap	PO1, PO2, PO3, PO4, PO5; PSO1	SDG 3, SDG 9, SDG 12	
3	Herbal Sanitary Napkins	Development of an eco-friendly herbal hygiene product using natural materials and Spirulina-associated components	Herbal sanitary napkin prototype/product	PO3, PO4, PO5, PO7; PSO1	SDG 3 – Good Health and Well-being; SDG 12 – Responsible Consumption and Production	
ONGOING RESEARCH ACTIVITIES						
4	Spirulina-Based Bioethanol Production via Bioconversion	Bioconversion of Spirulina biomass for production of bioethanol as a renewable energy source	**Spirulina-derived bioethanol** through biomass bioconversion	PO1, PO2, PO3, PO4, PO7; PSO2	SDG 7, SDG 9, SDG 12, SDG 13	
5	Electricity Generation via Spirulina Algae in a Microbial Fuel Cell System	Investigation of **Spirulina algae-based microbial fuel cell technology** for electricity generation and sustainable energy recovery	**Spirulina-based microbial fuel cell system** for electricity generation	PO1, PO2, PO3, PO4, PO7; PSO2	SDG 7, SDG 9, SDG 12, SDG 13	
6	Anaerobic Bioconversion of Residual Biomass to Biogas	Anaerobic conversion of residual algal biomass into biogas for energy recovery	**Biogas generation from residual Spirulina biomass** and resource recovery	PO1, PO2, PO3, PO4, PO7; PSO2	SDG 7, SDG 9, SDG 12, SDG 13	
7	Phycocyanin-Integrated Hydrogel for Wound Healing Applications	Incorporation of Spirulina-derived phycocyanin into a hydrogel platform for wound-healing applications	**Phycocyanin-integrated hydrogel formulation/prototype** for wound-healing studies	PO1, PO2, PO3, PO4, PO5; PSO1	SDG 3, SDG 9	

8	IoT-Enabled Real-Time Biorefinery Monitoring & Process Control	Integration of IoT-based sensing and monitoring for real-time observation and control of parameters during Spirulina cultivation and biorefinery processes	**IoT-enabled monitoring and process-control system** for real-time tracking of biorefinery parameters	PO1, PO2, PO3, PO4, PO5, PO9, PO11; PSO1, PSO2	**SDG 9, SDG 12, SDG 13**	
9	Spirulina-Mediated Wastewater Remediation & Nutrient Recovery	Utilization of Spirulina for treatment of wastewater and recovery of nutrients through algal biomass cultivation	**Spirulina-based wastewater remediation and nutrient recovery system with treated/purified water as the process output**	PO1, PO2, PO3, PO4, PO7; PSO1, PSO2	**SDG 6, 12, 13**	

7.5.2 INDUSTRY-SUPPORTED BIOMEDICAL RESEARCH LABORATORY

Research Focus - Biomedical Signal Acquisition, Human Movement Analysis and IoMT-Based Healthcare

The Industry-Supported Biomedical Research Laboratory, supported by Prashan Medical Technologies, provides specialized facilities for application-oriented research and student projects in biomedical instrumentation. The facility supports human movement and gait analysis, IoMT-based healthcare systems, EMG and EOG signal acquisition, enabling students to develop and evaluate biomedical prototypes for rehabilitation, physiological monitoring and assistive healthcare applications.

Table 7.5.2.1 Facilities and Research Application

Facility / System	**Research Application**
Motion Capturing System	Human movement and gait analysis
IoMT Kit	IoT-enabled biomedical monitoring and connected healthcare applications
EMG Acquisition System	Acquisition and analysis of electromyographic signals
EOG Acquisition System	Acquisition and analysis of electrooculographic signals

Figure 7.5.3 Industry Supported Research Laboratory

The Industry-Supported Biomedical Research Laboratory, supported by Prashan Medical Technologies, provides specialized facilities for research and student projects involving biomedical signal acquisition, human movement analysis and IoMT-based healthcare systems. The facility enables acquisition and analysis of EMG and EOG signals, gait and posture assessment using motion capture, and development of connected biomedical systems. It supports application-oriented research in rehabilitation, assistive healthcare, physiological monitoring and wearable biomedical systems, providing a platform for prototype development, testing and validation.

7.5.2.1 Industry Supported Laboratory – Prashan Medical Technologies, Coimbatore

S.No.	Project / Research Title	Research Focus / Application	Outcome / Output	Relevant PO/PSO	Relevant SDG	Evidence
	Vibro-Thermal Neurostimulation Wearable System for Phantom Pain Control	Development of a wearable system integrating vibro-thermal stimulation for management of phantom pain	**Wearable vibro-thermal neurostimulation prototype**	PO3, PO4, PO5, PO9; PSO1	SDG 3, SDG9	
ONGOING RESEARCH ACTIVITIES						
	Multimodal Sensor-Based FES Device for Assisted Lower Extremity Movement	Integration of FES with multimodal sensing for assisted lower-limb movement and rehabilitation	**Sensor-integrated FES prototype** for assisted lower-extremity movement	PO3, PO4, PO5, PO9, PO10; PSO1, PSO2	SDG 3, SDG 9	
	Adaptive Gamified sEMG Biofeedback System for Muscle Rehabilitation	Development of an adaptive sEMG-based biofeedback system that provides real-time feedback on muscle activity to support rehabilitation exercises	**Adaptive gamified sEMG biofeedback prototype for real-time muscle activity feedback and rehabilitation performance monitoring**	PO1, PO2, PO3, PO4, PO5, PO9, PO10; PSO1, PSO2	SDG 3, 9	

PART E: First Year faculty and financial Resources

(Data to be filled in for the first year course faculty and budget allocation and utilization)

E1. First Year Student-Faculty Ratio (FYSFR)

Table No. E1.1: FYSFR details.

Year	Sanctioned intake of all UG programs (S4)	No. of required faculty (RF4= S4/20)	No. of faculty members in Basic Science Courses & Humanities and Social Sciences including Management courses (NS1)	No. of faculty members in Engineering Science Courses (NS2)	Percentage= No. of faculty members ((NS1*0.8) + (NS2*0.2))/(No. of required faculty (RF4)); Percentage=((NS1*0.8) + (NS2*0.2))/RF
2023-24(CAYm2)	1320	66	71	53	102
2024-25(CAYm1)	1500	75	69	62	90
2025-26(CAY)	1620	81	48	48	59

E2. Budget Allocation, Utilization, and Public Accounting at Institute Level

Table No. E2.1: Budget and actual expenditure incurred at Institute level.

Items	Budgeted in 2025-26	Actual Expenses in 2025-26 till	Budgeted in 2024-25	Actual Expenses in 2024-25 till	Budgeted in 2023-24	Actual Expenses in 2023-24 till	Budgeted in 2022-23	Actual Expenses in 2022-23 till
Infrastructure Built-Up	1100.00	1065.73	950	919.83	410	390.96	485	470.79
Library	15	14.42	12	10.86	10	8.86	12	9.83
Laboratory equipment	235	221.99	102.5	101.84	88.5	83.87	72	68.82
Teaching and non-teaching staff salary	1850.5	1848.53	1800.00	1794.83	1550.00	1504.76	1450.00	1433.85
Outreach Programs	35.00	30.56	35	28.63	12	9.56	10	8.25
R&D	80	72.48	90	80.61	60	56.78	45	47.14
Training, Placement and Industry linkage	70	51.38	70	62.12	45	39.97	50	38.51
SDGs	45	37.98	45	39.9	25	22.18	25	17.3
Entrepreneurship	45	40.23	50	45.63	10	5.63	10	3.25
Others, specify	1350.5	1322.75	720.00	701.84	680.00	661.02	560.00	548.54
Total	**4826.00**	**4706.05**	**3874.50**	**3786.09**	**2890.50**	**2783.59**	**2719.00**	**2646.28**

E3. Budget Allocation, Utilization, and Public Accounting at Program Specific Level

Table No. E3.1: Budget and actual expenditure incurred at program level.

Items	Budgeted in 2025-26	Actual Expenses in 2025-26 till	Budgeted in 2024-25	Actual Expenses in 2024-25 till	Budgeted in 2023-24	Actual Expenses in 2023-24 till	Budgeted in 2022-23	Actual Expenses in 2022-23 till
Laboratory equipment	200000	175600	140000.00	129300.00	225000.00	219712.00	400000	410993.00

Software	0	0	0	0	0	0	0	0
SDGs	90000	85000	100000	99500	50000	58700	30000	26000
Support for faculty development	90000	89250	50000	47330.00	35000	32950.00	25000	18425.00
R & D	90000	87350	40000.00	46500.00	30000	27550	20000	15500.00
Industrial Training, Industry expert, Internship	90000	87550	40000	50500	50000	55550	15000	15750.00
Miscellaneous Expenses*	25000	25000	20000.00	15000.00	10000	5000	10000	10000
Total	**585000**	**549750**	**390000.00**	**388130.00**	**400000.00**	**399462.00**	**500000**	**496668.00**